

Macro Seasons

Data sources, season rules and monthly portfolio construction.

At each month-end, the model calculates one Growth score, one Inflation score and one Liquidity score. Growth and Inflation set the four season weights. Liquidity changes the final mix between risk and defensive ETFs.

1 The four seasons

Season	Growth and inflation	Economic reading	Fixed sleeve emphasis
SPRING	Growth rising Inflation cooling	Recovery with disinflation	Broad equities, technology, credit, housing and modest duration
SUMMER	Growth rising Inflation heating	Reflationary expansion	Cyclicals, value, energy, materials, emerging markets, commodities and TIPS
FALL	Growth falling Inflation heating	Stagflation	Gold, broad commodities, inflation protection, dollar, defensive equities and short duration
WINTER	Growth falling Inflation cooling	Disinflationary slowdown	Treasuries, defensive and low-volatility equities, gold, safe-haven currencies and cash

2 Data used for each pillar

GROWTH Data and source ALFRED: Industrial Production (INDPRO) and Nonfarm Payrolls (PAYEMS). FRED: Initial Jobless Claims (ICSA). Yahoo Finance: XLI, XLP, SPY and IEF adjusted prices. Changes used: industrial production over 6m; payrolls over 3m; claims over 3m (falling claims count as stronger growth); XLI versus XLP and SPY versus IEF over 6m.	INFLATION Data and source ALFRED: Consumer Price Index (CPIAUCSL). FRED: Five-Year Breakeven Inflation (T5YIE) and WTI crude oil (DCOILWTICO). Yahoo Finance: TIP and IEF adjusted prices. Changes used: change in 12-month CPI inflation over 3m; breakeven inflation over 3m; WTI and TIP versus IEF over 6m.	LIQUIDITY Data and source ALFRED: M2 money stock (M2SL). FRED: NFCI, high-yield OAS, 2-year Treasury yield, Fed balance sheet, reverse repos and the Treasury General Account. Changes used: NFCI level and 3m change; OAS 3m; M2 6m; 2-year yield 12m; Fed net liquidity 3m. Portfolio effect: easier liquidity can move up to 15 points from defensive ETFs to risk ETFs; tighter liquidity can move up to 25 points the other way.
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3 From month-end data to next month's portfolio

1	Select the month-end data	Use ALFRED decision-date values for CPI, industrial production, payrolls and M2. Apply the assigned publication delay to the other FRED series, which use latest-revised history. Use Yahoo adjusted prices through month-end.
2	Calculate three scores	Calculate the listed trailing changes and set their signs so positive means stronger growth, higher inflation or easier liquidity. Standardize each reading against up to 120 prior months, with at least 36 months required and values capped at three standard deviations. Average the available readings and apply a two-month exponential average.
3	Calculate four season weights	Stronger growth raises Spring and Summer; weaker growth raises Fall and Winter. Higher inflation raises Summer and Fall; lower inflation raises Spring and Winter. The four values total 100% and are used directly as sleeve weights. They classify current conditions rather than forecasting event probabilities.
4	Build next month's holdings	Adjust each manually specified season template with trailing volatility, blend all four templates using the season weights, then apply liquidity, real-yield, credit, momentum, trend and portfolio-volatility rules in that order.
5	Record next month's return	Apply the month-end weights to the following month's returns. The long-only backtest charges 10 basis points per unit of turnover. The later physical L/S simulation applies its separate trading and financing assumptions.

4 Season portfolio templates

Strategic season sleeves Spring: SPY 20%, QQQ 15%, SMH 10%, IWM 10%, XHB 7.5%, HYG 15%, LQD 7.5%, IEF 10%, GLD 5%. Summer: XLE 15%, XLB 10%, XLI 10%, VLU 10%, IWM 10%, EEM 10%, DBC 12.5%, CPER 5%, GLD 7.5%, TIP 10%. Fall: GLD 17.5%, DBC 10%, XLE 7.5%, TIP 15%, UUP 10%, XLP 10%, XLV 7.5%, SHY 7.5%, BIL 7.5%. Winter: TLT 17.5%, IEF 17.5%, SHY 7.5%, GLD 12.5%, USMV 10%, XLP 10%, XLV 10%, FXY 5%, FFX 5%, BIL 5%.	How the sleeves are used ETF membership follows the economic exposures in section 1: growth, technology and credit in Spring; cyclicals and inflation hedges in Summer; real assets and defensives in Fall; Treasuries, defensives and safe-haven assets in Winter. The percentages are standing strategic starting weights; monthly returns do not re-estimate them. At each month-end, assets without enough price history are removed and the remaining weights are rescaled. Each sleeve is averaged with a trailing 36-month inverse-volatility version. The four sleeves are then combined using the current season weights before the liquidity and portfolio controls are applied.
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Portfolio construction and validation

How the long-only and L/S portfolios are built, tested and constrained.

5 Portfolio construction

LONG-ONLY PORTFOLIO

Weights sum to 100%. The portfolio holds long ETF positions and BIL. It has no shorts or borrowing.

1. Template adjustment. For each season, start from the strategic weights in section 4. Calculate trailing 36-month volatility, form a version that divides each starting weight by its volatility, and average that version 50/50 with the starting weights.
2. Season and liquidity blend. Combine all four adjusted templates using the Growth/Inflation season weights. The Liquidity score can move up to 15 percentage points from defensive ETFs to risk ETFs, or up to 25 points from risk to defense.
3. Real yields and credit. The six-month change in the 10-year Treasury yield less five-year breakeven inflation rotates up to half of the available GLD or TLT/IEF sleeve. Credit stress turns on when high-yield OAS, or the earlier-history Baa yield spread proxy, exceeds 110% of its trailing 36-month median and has widened over three months; the model then halves the risk sleeve.
4. Momentum and trend. Rank current non-BIL holdings by return from 12 months ago to one month ago; the rank multiplier ranges from 0.75 to 1.25. Move each ETF below its 200-day moving average to BIL, using the 10-month average only when daily history is insufficient.
5. Portfolio volatility and cost. Estimate the current weight vector's volatility from the trailing 24 monthly returns. Scale can range from 0.50 to 1.00 around the 10% target; unused weight goes to BIL. Charge 10 basis points per unit of turnover.

This is the funded portfolio shown in the allocation table.

L/S PORTFOLIO

The portfolio combines two macro-season streams with a 13-ETF trend stream. Duplicate and opposing positions are netted before trading and financing costs are applied.

1. Two season streams. The core stream uses the season blend and Liquidity overlay. The enhanced stream uses the complete long-only process. Each stream is separately scaled between 0.50 and 1.50 using volatility through the previous month.
2. Time-series momentum stream. The 13 ETFs are SPY, QQQ, IWM, EFA, EEM, TLT, IEF, GLD, DBC, UUP, FXY, FXF and HYG. Each is long when its trailing 12-month return exceeds BIL and short otherwise.
3. Momentum sizing. Use trailing 36-month volatility so a 10%-volatility ETF receives roughly one-thirteenth of the stream. Cap each position at 20% of the stream, then scale the stream between 0.50 and 1.50 toward 10% volatility.
4. Combine streams. Weight the three streams inversely to their trailing 24-month volatility, measured through the previous month. Scale the combined return stream between 0.50 and 1.50 toward 10% volatility.
5. Execution and costs. Add duplicate positions, cancel opposing exposures, reduce BIL before recording a USD debit, and apply the gross, net, short, asset and borrowing limits. Charge commissions, regulatory fees, slippage, margin interest and short-position costs.

Long-only and L/S returns are reported separately.

6 Walk-forward research design

MONTHLY WALK-FORWARD

At each month-end, calculate the signals and holdings with data dated on or before that decision. Apply those holdings only to the following month's returns, then record trading and financing costs.

CUT-OFF TESTS

Rerun the model through earlier cut-off dates. Signals, holdings and returns through each cut-off must match the same dates in the full run. Any mismatch fails the test.

RESEARCH WINDOWS

Portfolio rules and overlays were compared using returns through December 2018. The selected combination was then evaluated from January 2019 through May 2026. Both samples contain simulated monthly returns.

POINT-IN-TIME RERUN

The macro history was rebuilt with decision-date vintages for CPI, industrial production, payrolls and M2. The rerun through July 2026 and the L/S execution history remain retrospective simulations.

7 Risk limits and data notes

Long-only limits

ETF and BIL weights total 100%. Exposure can fall as low as 50% when volatility is high; the remainder goes to BIL. Exposure cannot exceed 100%. Shorts and margin borrowing are disabled.

L/S limits

Maximum gross exposure: 175%. Maximum net exposure: 150%. Maximum gross shorts: 35%. Maximum absolute weight per non-cash ETF: 25%. Maximum USD margin debit: 50%. IBKR calculates the final Portfolio Margin requirement on the live account.

Execution cost assumptions

Assumptions reviewed August 20, 2026: \$0.005 per share, a \$1 order minimum, US regulatory-fee rates, 1 basis point of slippage and the applicable margin and short-proceeds tiers. It assumes 1% annual borrow for every short ETF because historical borrow data is unavailable. Taxes and market impact are excluded.

Historical inputs

ALFRED decision-date histories are used for CPIAUCSL, INDPRO, PAYEMS and M2SL. The remaining FRED inputs use coded publication delays applied to latest-revised histories. Yahoo adjusted prices provide traded returns. Mutual funds and gold futures extend selected ETF histories before inception.

Liquidity data limitation

FRED reports reverse repos in billions of dollars and the Fed balance sheet and Treasury General Account in millions. The current historical calculation does not rescale reverse repos before subtraction, which understates that term. Any correction will change the historical liquidity readings, so previous and corrected results should be reported separately.